(1)

Parallel and Distributed computing

Lecture # 01

To perform operation

Arithmetic — Logic

Perform → part computer

CPU

Main / Central — processing — unit

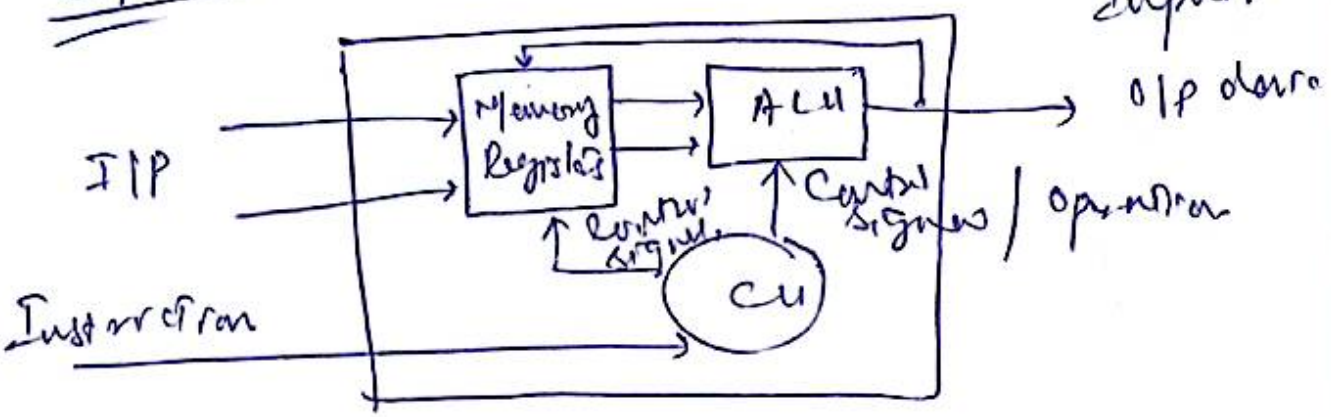
Sub units

Arithmetic — Logic

ALU → Perform operation on data

Input.

CPU



(2)

Input $\xrightarrow{\text{direct}}$ ALU ?

Input $\xrightarrow{\text{I/O device}}$ ALU ?
Indirect

Direct Study.

User 2 + 3 (NOT acceptable)

2 → ALU ← 3

↓ Result

Scenario.

i → Enter '2'

ii → Enter '3'

[After some time]

*: delay 2 is in the system and waiting for 3.

ALU Response

2 → ALU → Result

Garbage → ALU

ALU → Combinational circuits

O/P → I/P

Indirect Study

Input → Memory → ALU

⇒ ALU operation

↓

Control

↓

Control unit

(3)

computing

purpose → Fast computing

- Parallel → Start at same time, End at same time
- Distributed

→ parallel lines

- **Parallel** :- Parallel operation → ALU mode (parallel)

Memory → data

ALU → Operation End.

dependent on CU

1 2

parallel

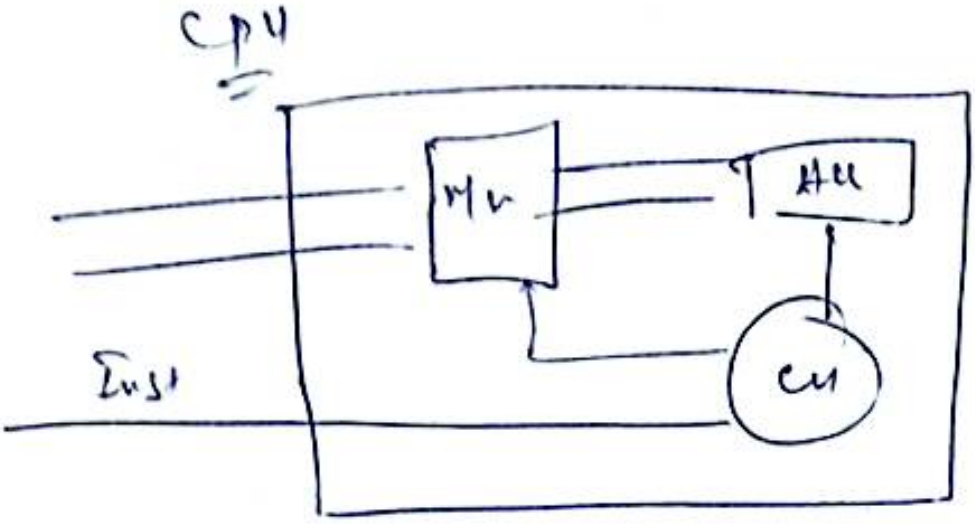
Array CU → parallel

Signal

for Memory

for ALU

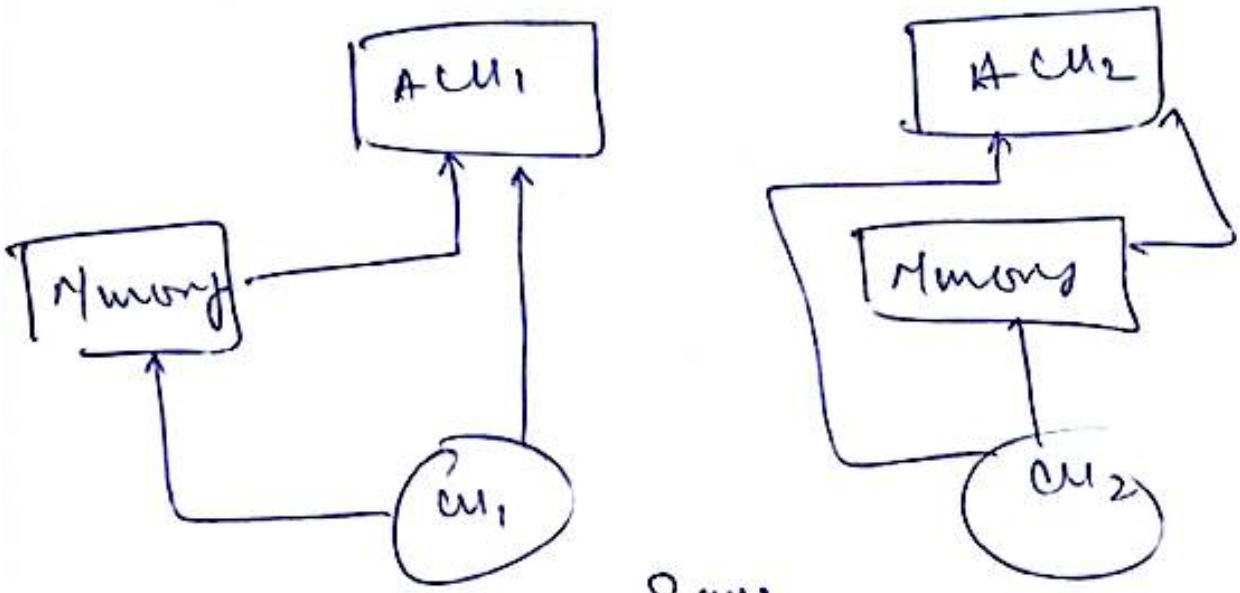
(1)



Key concept

i → 2 ALU → purpose parallel

- 2 - control signals
- 2 - Memory → data.

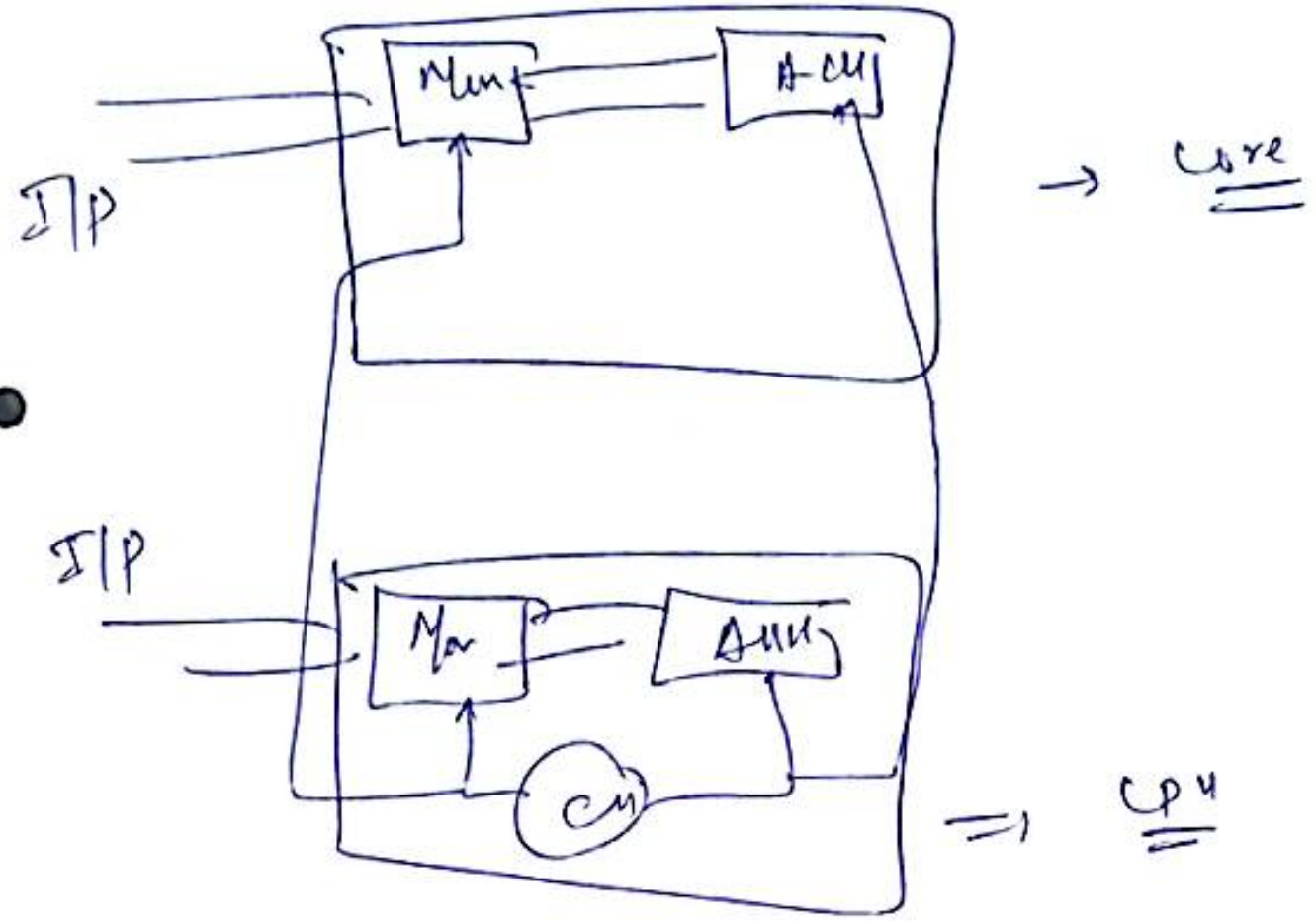


Same Instruction shared by

1 person — 2 person ?

5.

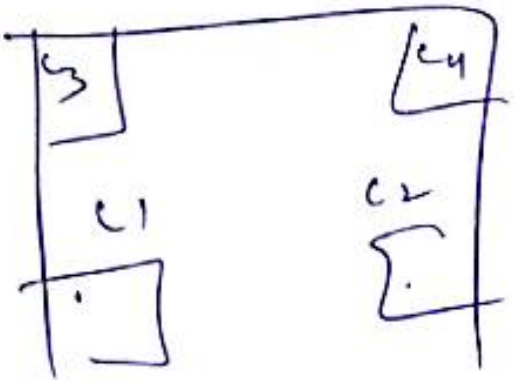
One CU required for parallel processing



- 2. control unit

1- CPU

1- computer



- parallel computing depend.
- CPU size → # of cores.

(6)

Distributed

↓

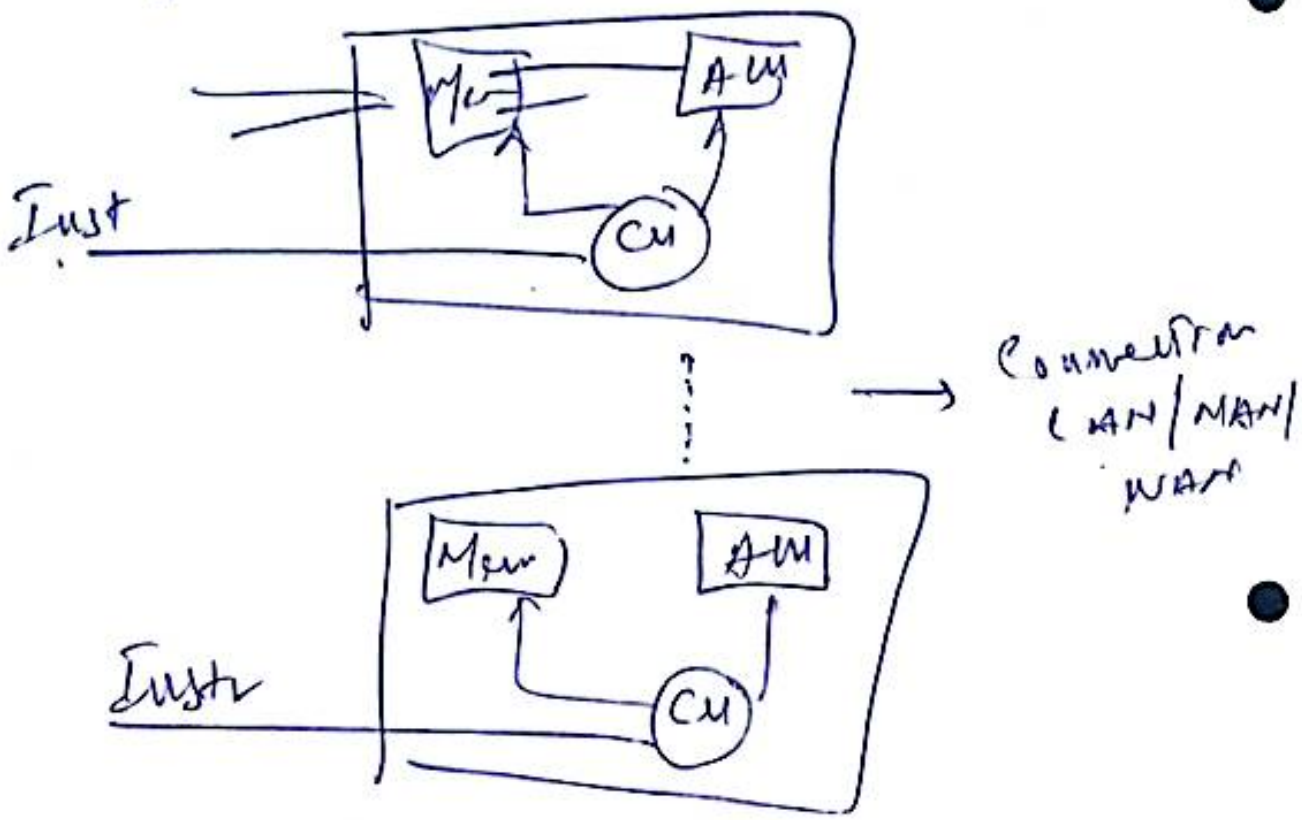
More than one computer

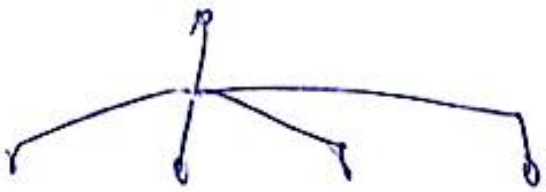
More than one process

↓

One C.U

CPU





(7) Flynn's Classification

Different classification of Computer Architecture.

- Inst Stream →
- Data stream.

Inst stream:

Inst → CU —C.S→ P.U.

Sequential flow

Mem

Data Stream

Inst Mem

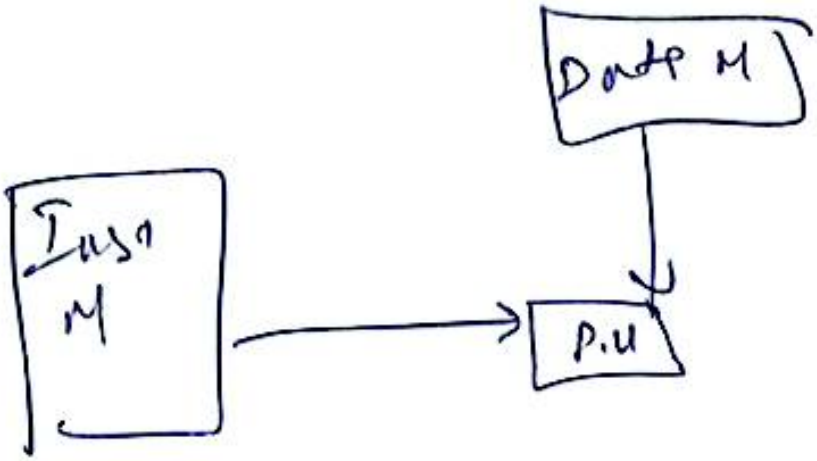
Data Mem

~~Data~~

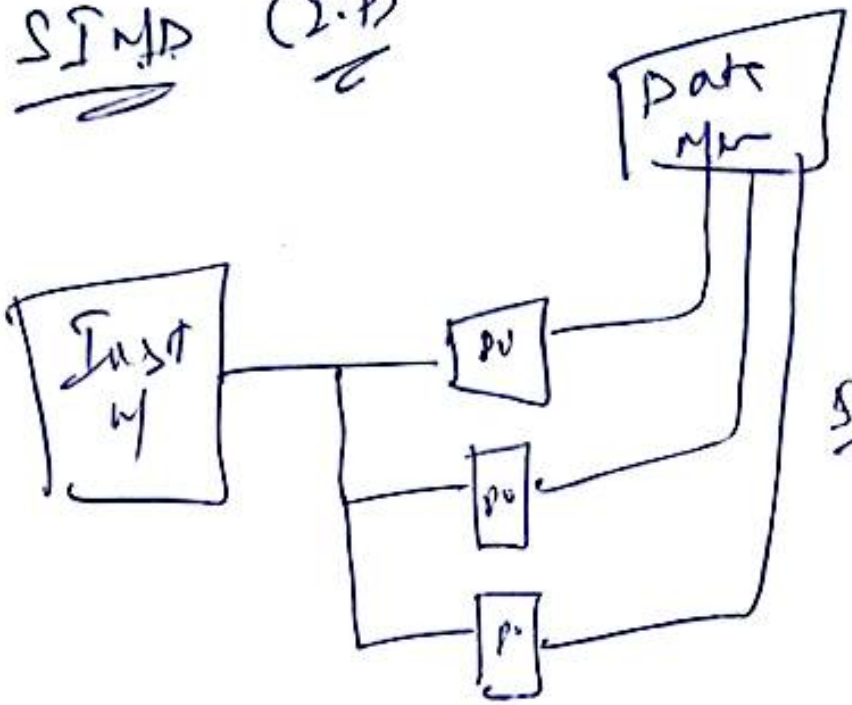
	Single	Multi
Inst	SISD	SIMD
Multi	MISD	MIMD

SISD

Load A
Load B
A = B + C
store A

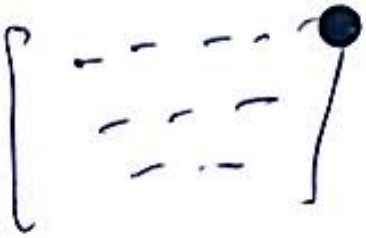


SIMD (I.P)



e.g
=
Image processing

SIMD
= 65X

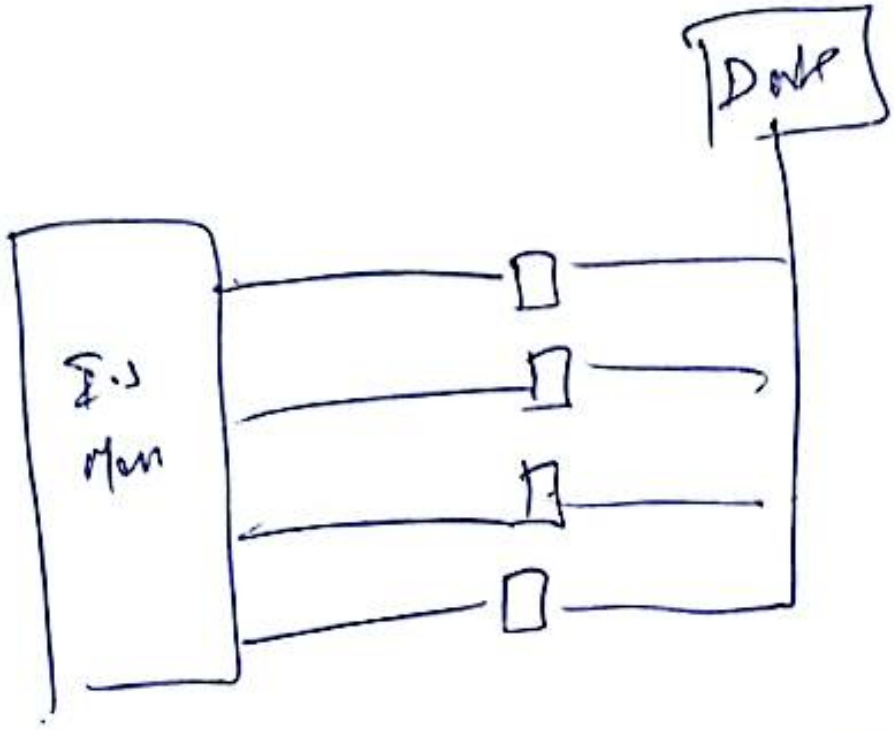


A(1) = B(1) + C(1)	A(2) = B(2) + C(2)

MISD

Same data set
&
different Algo

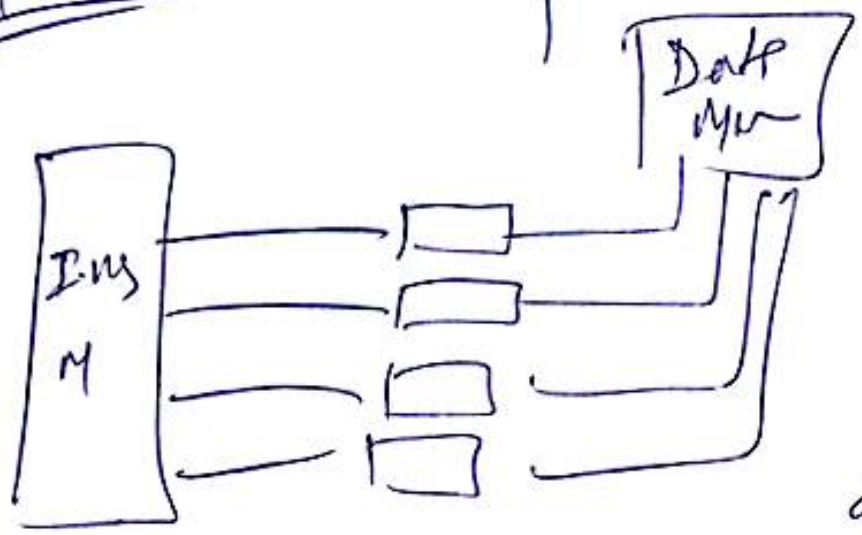
- Special computer
- task oriented



A = B(1) + C(1)	A = B(1) * C(1)

MIMD

DIstributed



i	2	3	4
+	-	*	/
d_1	d_2	d_3	d_4